Micah Robinson

Frisco, TX (**U.S. Citizen**) | (802)-337-6430 | mjrob.sys@gmail.com | linkedin | github | website portfolio

Education

The University of Texas at Dallas Richardson, TX
Bachelor of Science in Computer Science (Junior) *Aug. 2026 – May 2028*

- **Academic Path:** Planning M.S. in Computer Engineering, Aug 2027 (Focus: Embedded Systems & Avionics).
- **Relevant Coursework:** Computer Science II (C++), Discrete Mathematics I, Linear Algebra, Professional & Technical Communication.

Collin College Frisco, TX
Associate of Science in Computer Science (GPA: 3.88 / 4.00, Magna Cum Laude) *Aug. 2024 – May 2026*

- **Honors & Awards:** Phi Theta Kappa Honor Society, UTD Academic & PTK Transfer Scholarships.
- **Foundational Prep:** Calculus I & II (A grades), University Physics I (A grade) & II (Calculus-based).

Technical Skills

Languages: C, C++, Python, Java, JavaScript, HTML/CSS, Lua
Embedded Systems & Protocols: Real-Time Operating Systems (RTOS), I2C, SPI, UART, Register-Level Drivers, HAL, State Machines, Interrupt Handlers (ISR)
Hardware & Modules: STM32 Microcontrollers, ATmega328P (Arduino), MPU6050 IMU, BMP280 Barometer, SSD1306 OLED, Logic Analyzers, Multimeters
Developer Tools: VS Code, MSYS2 MinGW-w64 (g++), Git/GitHub, NetBeans, Linux/Bash

Embedded Systems & Software Projects

Custom STM32 Bare-Metal RTOS | *C, ARM Cortex-M, GCC, Git* Oct. 2026 – Present

- [**In Active Development**] Architecting a preemptive microkernel/RTOS targeting STM32 ARM Cortex-M microcontrollers to handle concurrent task scheduling and deterministic context switching.
- Implementing SysTick timer interrupts and PendSV assembly routines for atomic register context saving ($r4$–$r11$).
- Building priority-based thread management, Task Control Blocks (TCB), stack allocation, and inter-process synchronization primitives (Mutexes, Semaphores).

I2C Multi-Sensor & Telemetry Pipeline | *C++, ATmega328P, I2C, Logic Analyzer* May 2026 – Aug. 2026

- Authored custom register-level C++ drivers for MPU6050 6-DOF IMU, BMP280 barometer, and SSD1306 OLED display without third-party library abstraction wrappers.
- Implemented factory calibration math in software to calculate accurate real-time altitude, pressure, and temperature data.
- Analyzed and debugged I2C bus transactions, clock stretching, and signal timing using a hardware logic analyzer.

Arduino State-Machine "Simon Says" System | *C++, Hardware Interrupts, Shift Registers* | May 2026 – June 2026

- Designed a memory-efficient, deterministic state-machine architecture in C++ utilizing hardware interrupts for non-blocking button inputs.
- Authored a custom Hardware Abstraction Layer (HAL) to decouple game state logic from physical pin I/O.

Leadership & Academic Experience

Physics Laboratory Team Lead & Tutor Collin College, TX
Lead Lab Researcher & Academic Peer Tutor *Aug. 2025 – Dec 2025*

- Led a 4-person physics research group to a 98% average across advanced mechanics and electromagnetism experiments.
- Tutored peers in calculus-based physics and linear algebra concepts, improving coursework performance and problem-solving rigor.